Resistive, Inductive and Capacitive Transducer

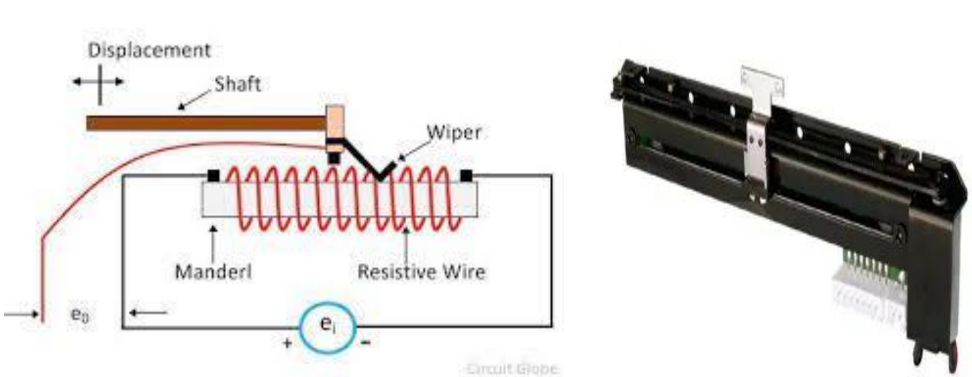
Dr. Sushma Gupta
Professor
Department of Electrical Engineering
MANIT, Bhopal

Resistive Type Transducer

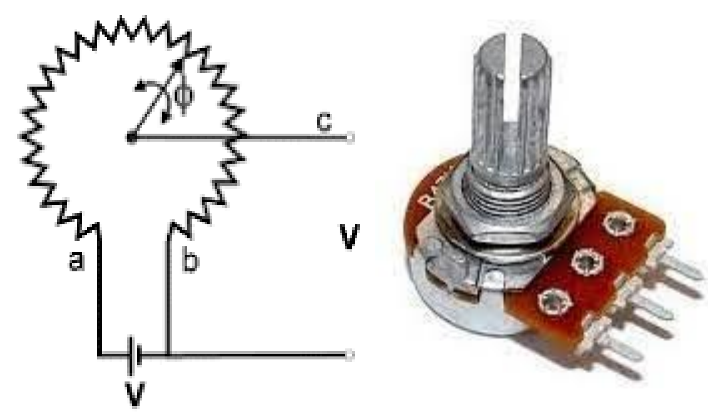
- **Resistance** of a **metal conductor** is expressed by a simple equation- x_t x_o

  $R = \rho L/A$

- Where ρ = Resistive of conductor material (Ω-m)
  - L= Length in (m)   A = Area in (m^2)
- If **any parameter** (L, A and ρ) is **changed** then **resistance will change.**
- The **translational**, **rotational** and **helical potentiometers** work on the principle of **resistance change** with **change in length**.



Translational Potentiometer

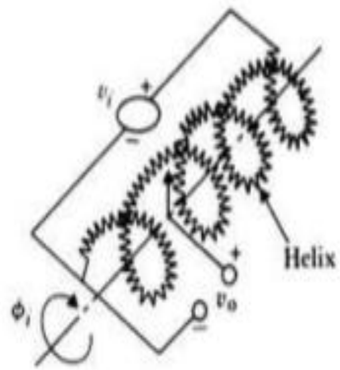


Rotational Potentiometer

- The output voltage under ideal condition is-
- V_o = [(Resistance at the o/p terminal) X Input Voltage]/ Resistance at the input terminal

$$e_o = X_o e_i / X_t$$

- For rotational potentiometer-
- $$V_o = \Phi_o V_i / \Phi_t$$
- Sensitivity = ***Output/Input***
- $$= V_o / V_i = X_o / X_t = \theta_o / \theta_t$$
- The ***resistivity of metal*** is ***changed*** with ***change in temperature*** thus causing ***change in resistance.***
- For **translational devices**, the **resolution is** limited to **25-50µm**.
- Variation in resistance is **in step** so the **resolution is also limited**.
- Actual practical limit is 20 to 40 turns per mm.
- For rotational devices, angular resolution is-
  - 3 to 6 degree/D Where D = Diameter of the potentiometer
- In order to get **high resolution thin wire** should be used and placed **very closely.**
- In case for **fine resolution and high resistance, Carbon-film** or a conductive-plastice resistance elements is used.
- **Resolution can be increased** by using **multi-turn potentiometer** which is known as **helipot or helical potentiometer.**

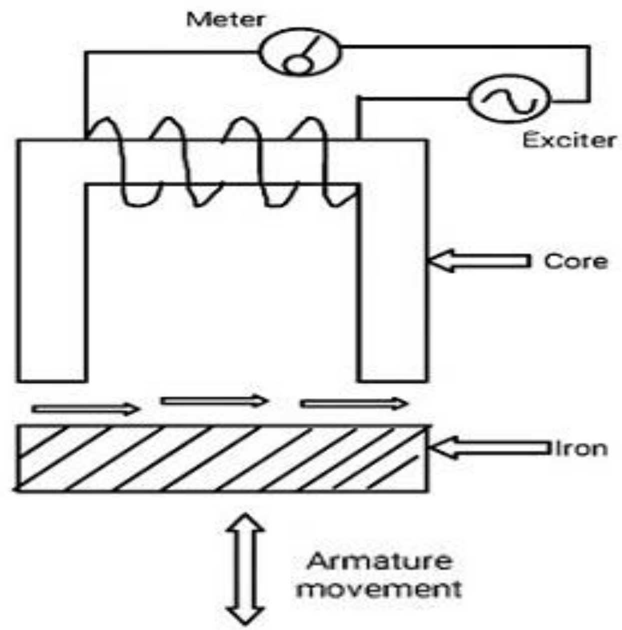


Helical Potentiometer

Inductive Transducer

- A **transducer** that works on the principle of **electromagnetic induction** is called an **inductive transducer**.
- A **self-inductance** or **mutual inductance** is varied to **measure required** **physical quantities** like displacement (rotary or linear), force, pressure, velocity, torque, acceleration, etc.

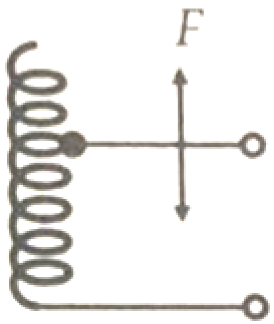
- ***Inductive transducers*** are divided into ***two categories-***
  - **Single Coil Inductance Type** **or** **Self-Inductance Type**-When the ***armature*** of the circuit is ***moved***, the **air gap** between the **core** **(iron, amorphous steel, ferrous ceramics)** and **iron** is **changed** therefore the **permeability** of the **flux** produced in the circuit is **changed**. This **results** in a **change of the inductance** in the circuit.



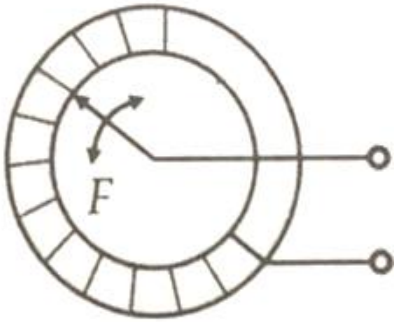
- **Self-inductance of the coil** can be expressed as-
- $L = N^2/R$ (1)
- N = Number of Turns, R = Reluctance of the magnetic circuit
- The **expression** for the **reluctance of the coil** is-
- $R = l/\mu A$ (2)
- Where l represents the length of the coil
- μ represents permeability of the coil,
- A represents a cross-section area of the coil,
- **Put the value of R from equation (2) into Eq. (1)**
  $L = N^2\mu A/l$
- $L = N^2\mu G$ (3)
- $G = A/l$ = geometric form factor
- From the **equation (3)**, it is concluded that **self-inductance** can be varied or changed by changing the **number of turns**, or **geometric form factor** ***(Cross sectional area and length)*** or **permeability of the coil**.

Inductor Coil Configuration

- ***Inductance*** can be changed with ***change in number of turns*** of coil. This arrangement is used to ***measure displacement***.



Linear displacement



Rotational displacement

- **Coil may be used as air cored or iron-cored.**
- ***Air cored coils*** -
  - **High frequency operation** because of absence of eddy-current losses.
  - **Inductance of air-cored coil** is independent of **coil current** as **permeability** of the **air** is **low**.
- **Size is large** compared to **iron cored coil**.
- **Change in inductance** of air core coil is **small** because of the **low permeability of air.**
- It is **affected** by **external magnetic field**.

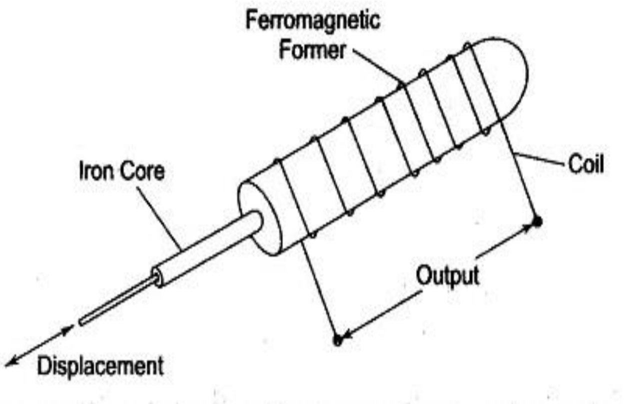
Air-cored coil

- **Iron or ferromagnetic cored coils are preferred-**
- The size of coil is small.
- This type of coil is less affected by external magnetic field because their magnetic field is confined to the iron core.
- Change in inductance is high due to high permeability of iron core.
- At high frequency, eddy current losses are high. Due to this reason supply frequency is kept below 20 kHz.
- Inductance of iron core coil depends upon the value of current carried by the coil.

Magnetic Core Coil

- Variation in geometric configuration
  - An air cored coil is divided into two parts .
  - One coil is fixed and another coil is movable.
  - The movable coil is attached to the object whose parameter has to be measured.
  - When parameter is changed, length of the coil is changed as a result geometric configuration is changed.
- Change in permeability-
  - An Inductive Transducer also works on the principle of the variation of permeability causing a change in self inductance.
  - The iron core is surrounded by a winding.
  - If the iron core is inside the winding, its permeability is increased, therefore inductance increases.
  - When the iron core is moved out of the winding, the permeability decreases, resulting in a reduction of the self inductance of the coil.
  - This transducer can be used for measuring displacement.

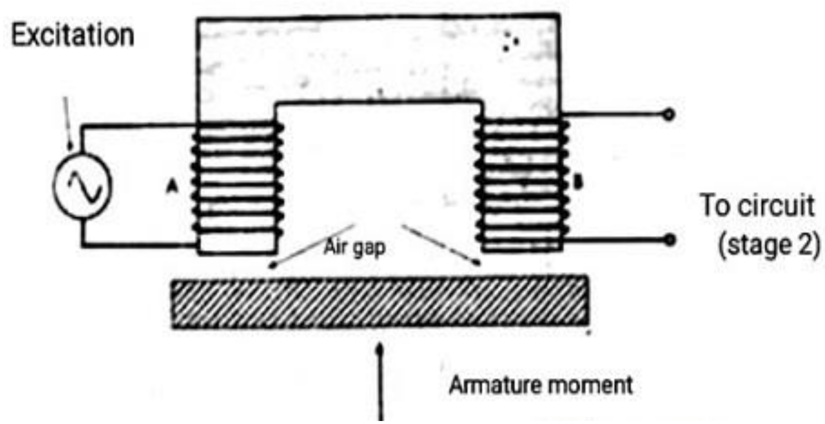
Geometric Configuration

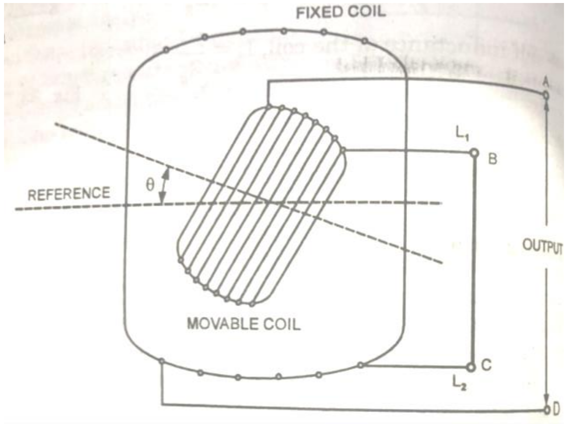


Variation in inductance with change in permeability

Change in Mutual Inductance

- ***Mutual Inductance phenomenon happens between two neighboring coils.***
- The first coil is fed with AC voltage and it generates flux.
- ***The same flux links to the second coil and induces emf in it.***
- The emf generated in the second coil is called the **mutually induced emf** and the phenomenon is called as **mutual inductance**.
- ***The mutually induced emf depends on the movement of the armature.***



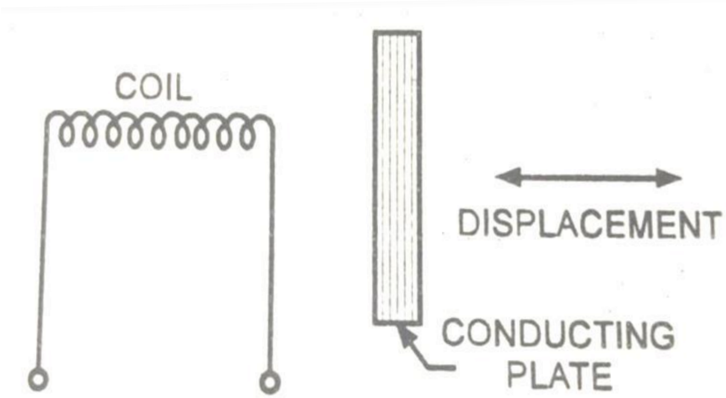


- When the armature position is changed by connecting to the movable mechanical element, then the **air gap** between the armature and the magnetic material is changed so the inductance changes.

- ***The mutual inductance of the coils depends upon self-inductance of coils (L_1 and L_2) and expressed as-***
- $$\mathbf{M = K \sqrt{L_1 L_2}}$$
- Where 'K' represents the coefficient of coupling.
- ***Hence, the mutual inductance can be changed by varying the self-inductance of the individual coils or by changing the coefficient of coupling.***
- The factor K depends on the distance and orientation of the coils.
- ***To measure the displacement, armature is connected to a movable object. As the object moves, the coupling factor K changes, which results in a change in mutual inductance in the coils.***
- This change in mutual inductance or output voltage can be calibrated in terms of displacement for an instrument.

Eddy Current Production

- When the conductive plate is placed near the coil that carries alternating current, ***circulating current*** is induced in the ***conducting plate*** which has its ***own magnetic field*** acts ***against*** the ***coil's magnetic field***.
- ***The conductive plate that carries circulating current is called eddy current.***
- When the conductive plate is brought near to the coil, then the eddy current is produced more therefore generates more magnetic flux in conducting plate, which reduces the magnetic flux of the coil and inductance.
- ***As the distance between the coil and the conductive plate is decreased, higher eddy currents are produced and more reduction in the inductance of the coil and vice versa.***
- Hence the change in inductance can be measured by moving the conductive plate. This Change can be calibrated to measure the physical quantity called displacement in an instrument.

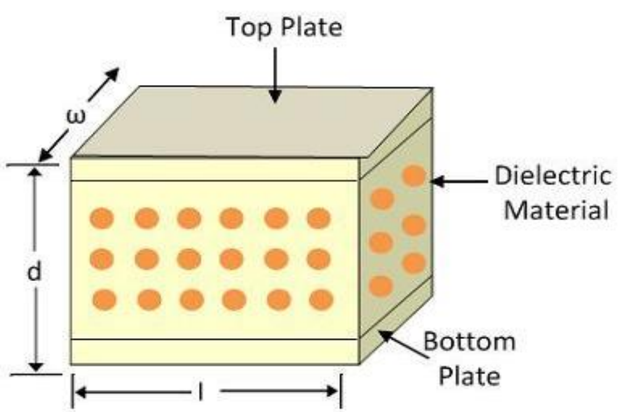


Capacitive Transducer

- The **working principle of a capacitive transducer** is that **capacitance varies** with **change in physical parameters**.
- As per its structure, **Capacitors** are having **two parallel metal plates** which are maintaining the **distance** between them.
- In between them, **dielectric medium** (such as air, material, gas or liquid) is to be **filled**.
- The **capacitance** of the **variable capacitor** can be measured by this formula.

$$C = \frac{\varepsilon_0 \varepsilon_r A}{d}$$

- C indicates the capacitance of the variable capacitance.
  ε_o indicates the **permittivity** of **free space**.
  ε_r indicates the **relative permittivity**.
  A indicates the **area of the plates**.
  D indicates the **distance** between the **plates**.

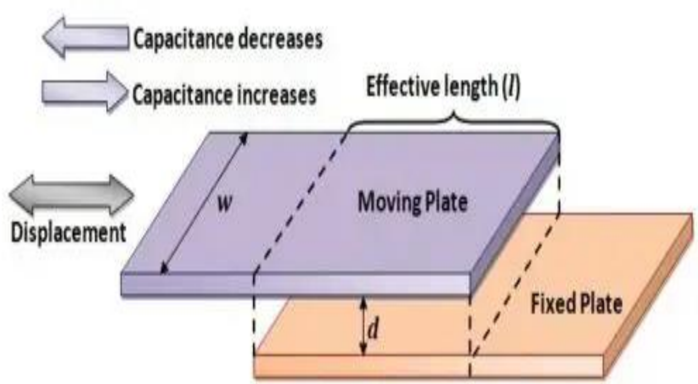


- It is a **passive type** of transducer means external power is required for operation.
- The capacitive transducer works on the principle of **change of capacitance** which may be caused by-
  - Change in **overlapping area**.
  - Change in the **distance** between the plates
  - Change in **dielectric constant**.
- **These changes** (Change in overlapping area and distance between the plates) are caused by **physical variable** like displacement, force and pressure.
- The **change in capacitance** with **change in dielectric constant** causes in measurement of **liquid and gas level**.
- The **change in capacitance** may be measured with the **Schering bridge**.

Change in Overlapping Area

- **Parallel Plate Capacitor with Rectangular or square Plates-**
- In this type of transducer, **one plate is fixed** and the **other plate can be moved**.
- **Physical parameter** variations are **applied** at **movable plate** as a result **overlapping area A** will **vary**.

- As a results, **capacitance** value of this transducer will change.
- $C = \varepsilon_o \varepsilon_r (w\underline{L})/d$



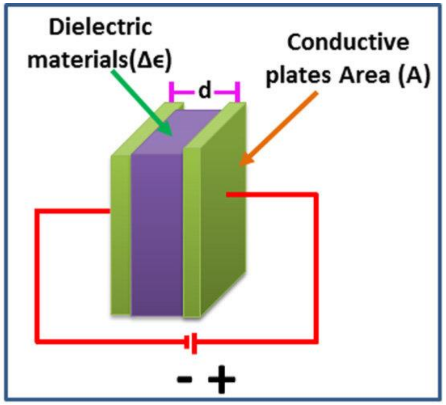
Where

L = length of overlappin part of the plates.

w = Width of the plates.

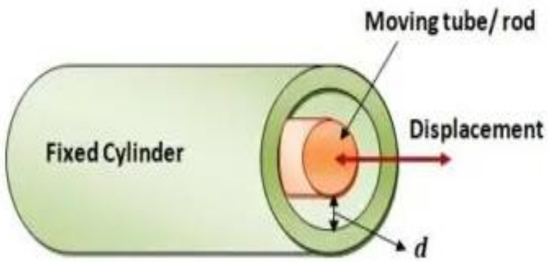
d = Distance between these two plates

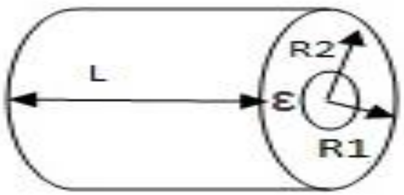
$\varepsilon_o \varepsilon_r$ = Free space and Relative permittivity



- <u>**Cylindrical Capacitive Transducer-**</u>
- L – Overlapping length of the cylinder
- R_2- Inner diameter of outer cylinder
- R_1 – Outer diameter of inner cylinder
- The capacitance

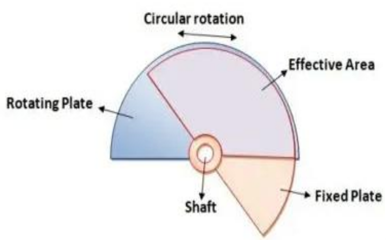
$$C = \frac{2\pi\varepsilon_o\varepsilon_r L}{\log(R_2/R_1)}$$





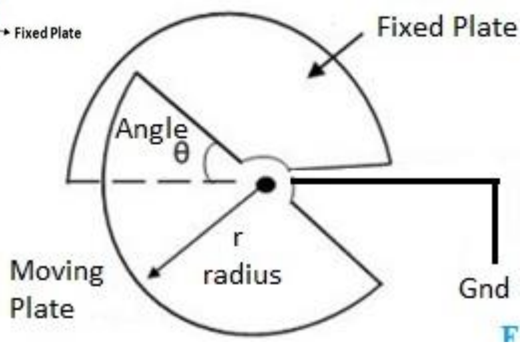
- <u>Semi Circular Capacitive Transducer for angular displacement-</u>
- There are two plates. One plate is fixed and another plate is movable.
- The angular displacement to be measured is applied to movable plate.
- The angular displacement changes the overlapping area of the plates and thus changes the capacitance.

$$C = \frac{\varepsilon A}{d} = \frac{\varepsilon\theta r^2}{2d}$$



- The capacitance is maximum when the two plates completely overlapping to each other $\theta = 180^o$

$$C = \frac{\varepsilon A}{d} = \frac{\pi\varepsilon r^2}{2d}$$



Sensitivity of Capacitance

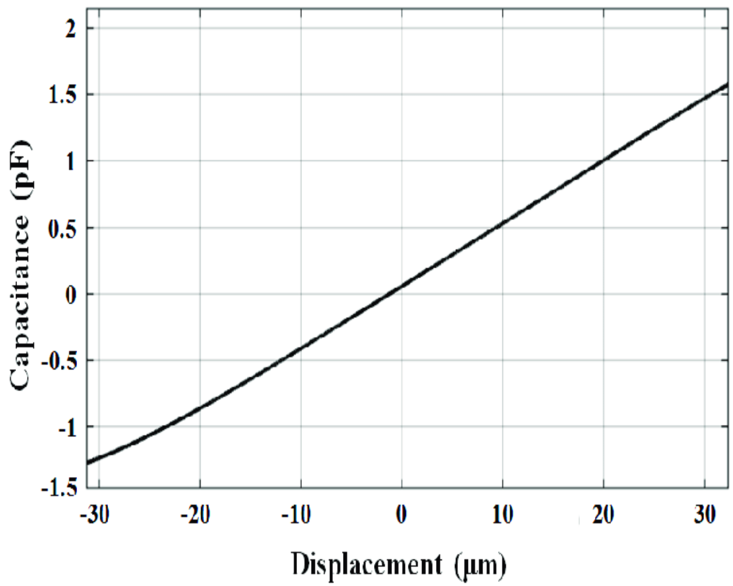
- Sensitivity is defined as change in capacitance with change in length-

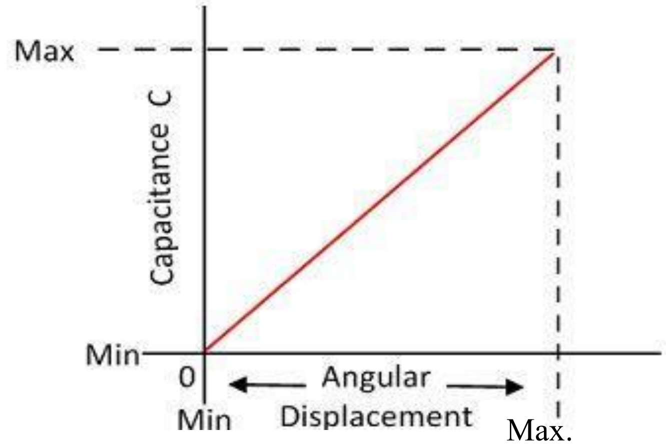
$$S = \frac{\partial C}{\partial L} = \varepsilon \frac{w}{d} \ \text{F/m}$$

- So the **sensitivity is constant**, and therefore it gives the **linear relation** between the **capacitance** and **linear displacement** (Overlapping Area).

- The sensitivity for **angular displacement**-

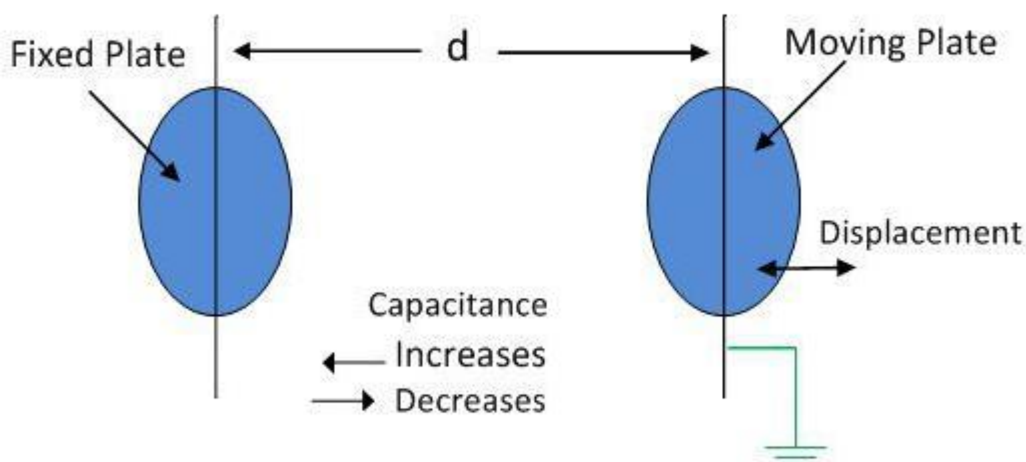
$$S = \frac{\partial C}{\partial \theta} = \frac{\varepsilon r^2}{2d}$$

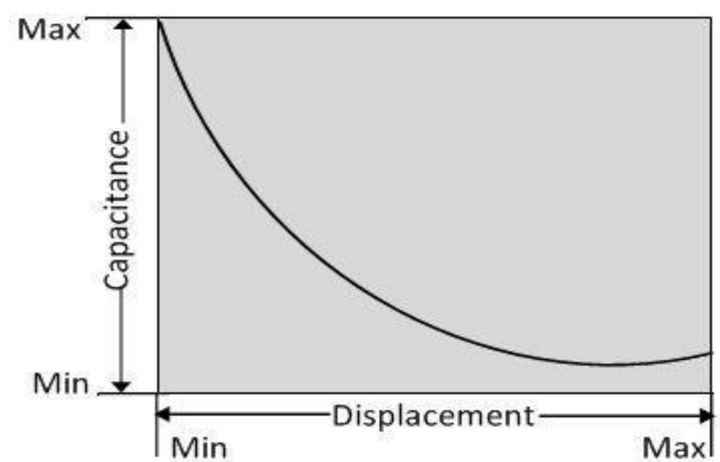




Transducer using the change in distance between the plates

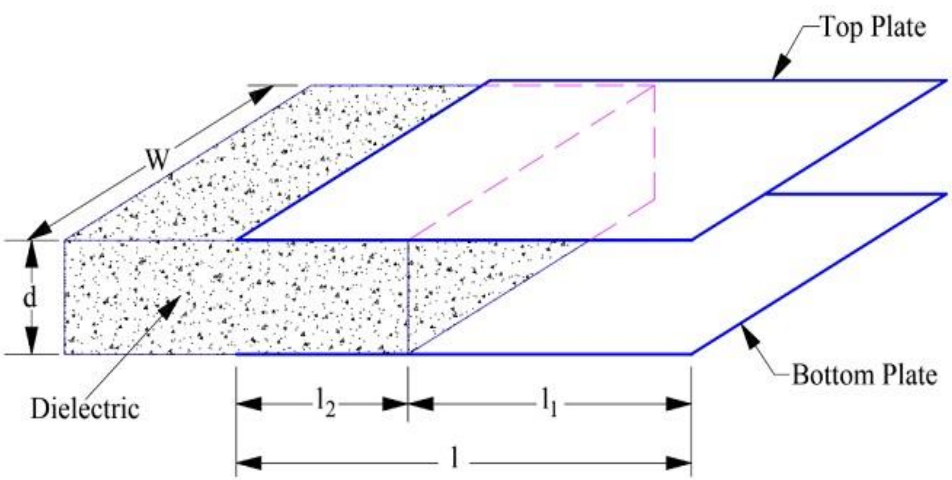
- The **capacitance of the transducer** is **inversely proportional** to the **distance** between **the plates**.
- The **one plate** of the transducer is **fixed**, and **other is movable**. The **displacement** which is to be measured *connected to movable plates*.
- The **capacitance** is **inversely proportional** to the **distance** therefore the capacitor shows the **nonlinear response**.
- Sensitivity is defined as- $S = \frac{\partial C}{\partial d} = \varepsilon \frac{A}{d^2} \; F/m$
- This type of transducer is used for measuring the **small displacement**.





Change in Capacitance with change in Dielectric Constant

- A **capacitive transducer** may use the principle of **change in dielectric constant** to achieve variable capacitance.
- **This principle is utilized** in capacitive transducer for the measurement of **linear displacement and level**.
- The **plates** of capacitor are **fixed**. A **moving object** having some **dielectric constant ε_r** is moving **into the plates**.
- At any **intermediate stage**, let the **l_2 length of object** is **inside** the plates. Therefore, up to **l_2 length**, the capacitor is filled up with **dielectric** having **dielectric constant ε_r** whereas **l_1** length have **air**.
- The capacitance in this combination can be found as-



$$C = \frac{\varepsilon_0 W}{d}[l_1 + \varepsilon_r l_2]$$

As the **object moves into** the capacitor, the value of **l_2 increases** and hence the capacitance **C increases**.

By **measuring** this **capacitance**, the **linear displacement** or **level** can be measured.

Advantages and Disadvantages of Capacitive Transducer

- It is very sensitive transducer.
- It has high input impedance so loading effect is less.
- It requires very small power for operation.
- It gives good frequency response because of which it can be used for dynamic study.
- Metallic parts of the transducer require insulation.
- Some transducer shows nonlinear characteristics so operation is in limited range.
- Proper earthling is required to reduce the effect of stray magnetic field.

- In a variable inductive transducer, the coil has an inductance of 2.5 mH when the effective turns on the coil are 50. Determine the inductance of the coil when the measurand makes the effective turns on the coil 52.
- A capacitive transducer with its plate separation of 0.05 mm, under static conditions has a capacitance of 5×10^{-12} F. Determine the change in displacement, which causes a change of capacitance of 0.75×10^{-12} F.
- A capacitive transducer is made up of two concentric cylindrical electrodes. The length of electrodes is 25 mm, the inner diameter of outer cylindrical electrodes is 4.2 mm and the outer diameter of inner cylindrical electrode is 4.0 mm. Determine the sensitivity of the transducer. Assume air medium. Determine the change in capacitance for a displacement of the inner electrode of 2.5 mm. Determine also the electric stress when a voltage of 150 V is applied across the electrodes.

- Inductance of Coil = 2.704 mH

- Change in Capacitance $\Delta C = 0.75 \times 10^{-12}$ F
- Unknown displacement= 0.333 mm

- Sensitivity = 1.14 pF/mm
- Change in capacitance = 2.8 pF
- Air gap length between two electrodes = (D-d)/2 = (4.2-4.0)/2 = 0.1 mm
- Dielectric Stress = V/ Air gap length = 150/0.1 =1500 V/mm